Non-regular languages

(Pumping Lemma)

Original Slides by C. Busch, LSU. Modified by A. H. Özer, Marmara University.

Non-regular languages

$$\{a^n b^n : \ n \geq 0\}$$

$$\{vv^R : \ v \in \{a,b\}^*\}$$

Regular languages

$a^* b$

$b^* c + a$

$b + c(a+b)^*$

etc...

How can we prove that a language L is not regular?

Prove that there is no DFA or NFA or RE that accepts/generates L

Difficulty: This is not easy to prove

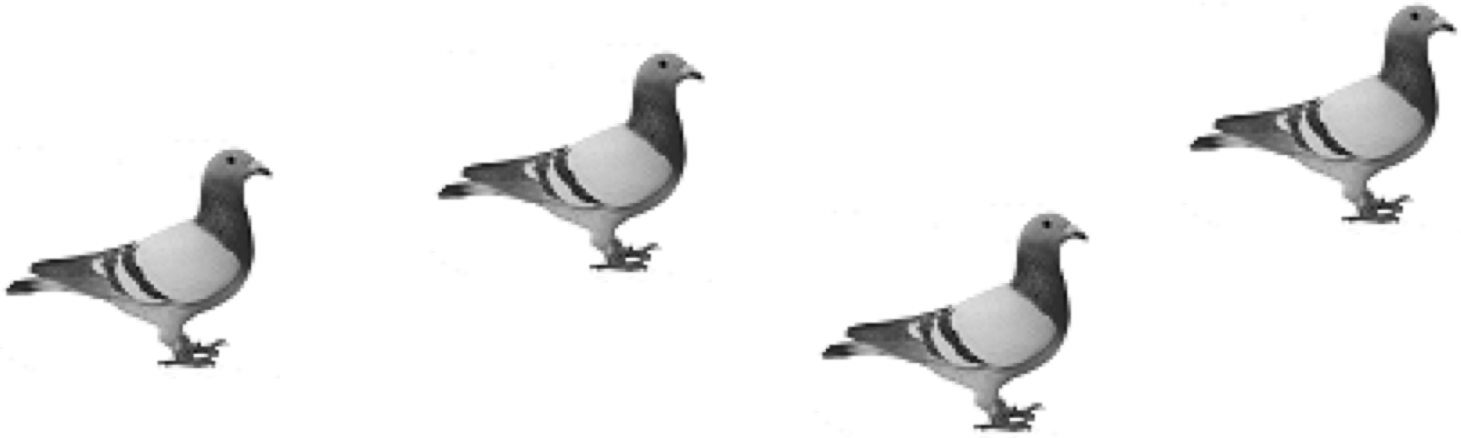
(since there is an infinite number of them.)

Solution: Use the Pumping Lemma!

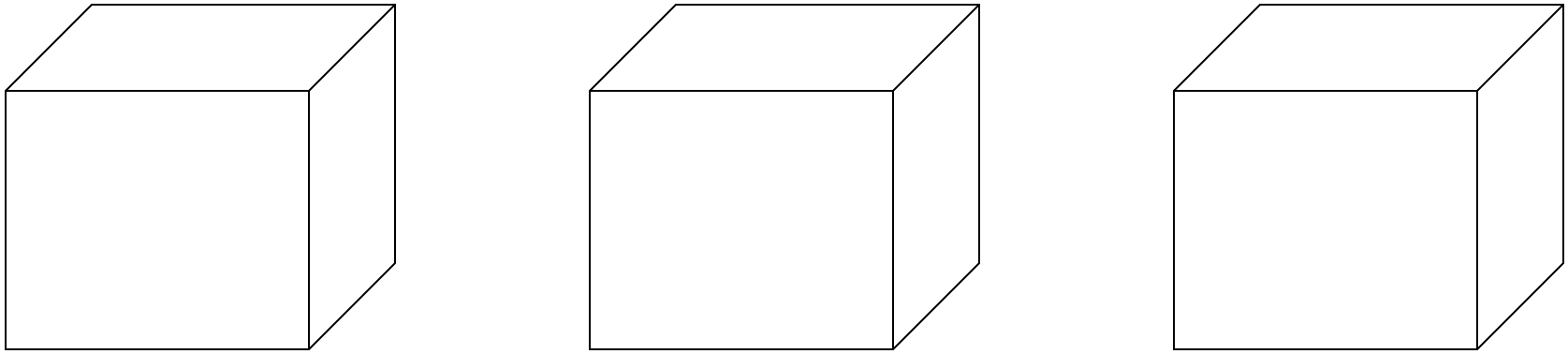


The Pigeonhole Principle

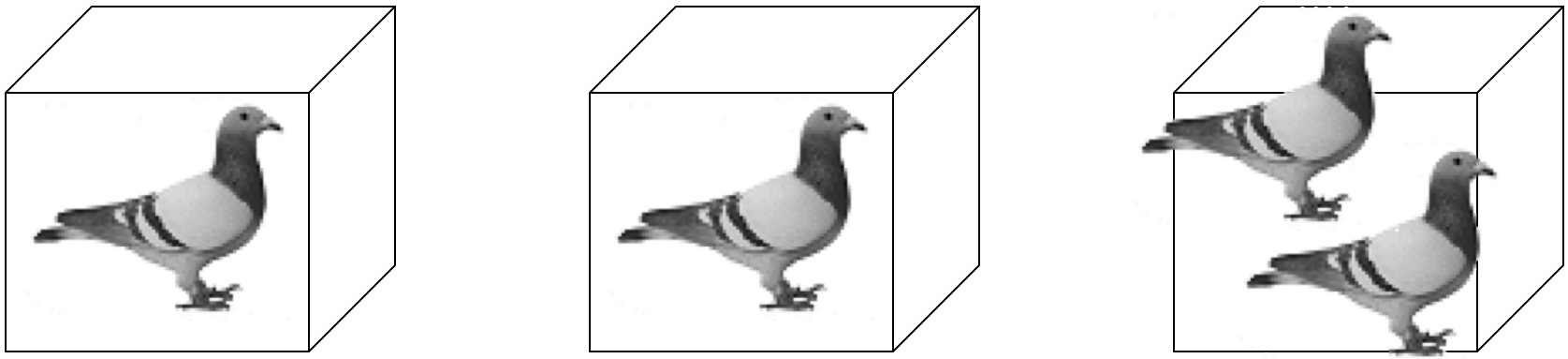
4 pigeons



3 pigeonholes



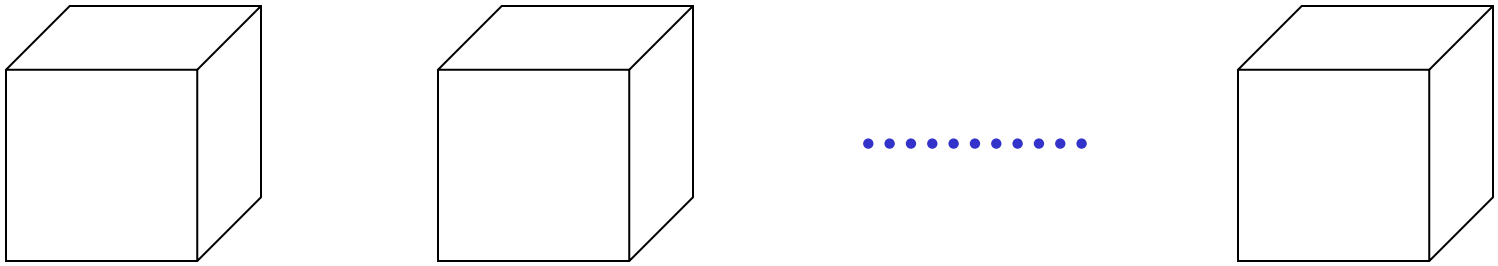
A pigeonhole must
contain at least two pigeons



n pigeons



m pigeonholes $n > m$



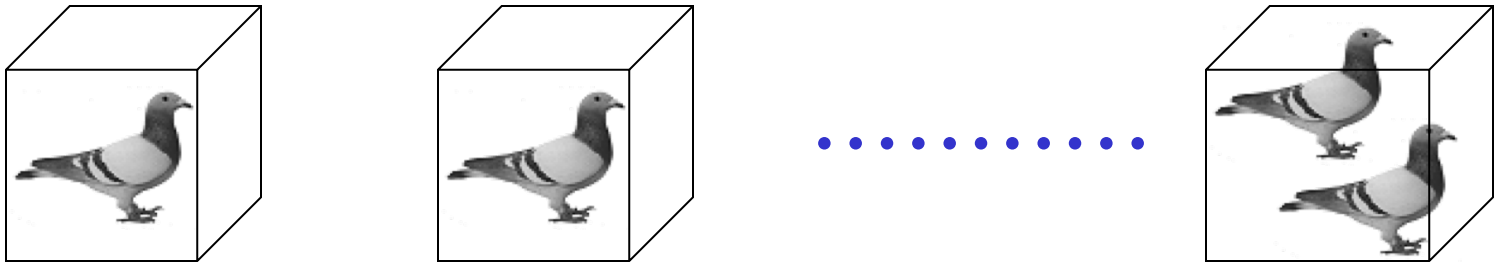
The Pigeonhole Principle

n pigeons

m pigeonholes

$n > m$

There is a pigeonhole with at least 2 pigeons

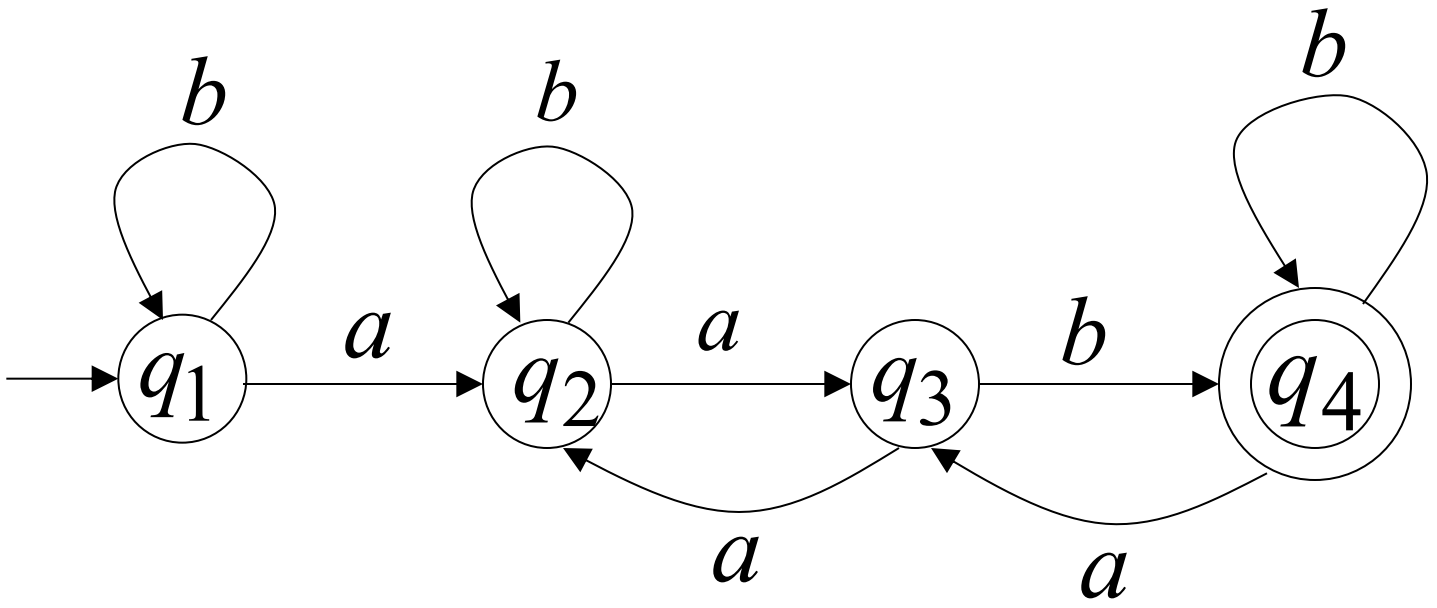


The Pigeonhole Principle

and

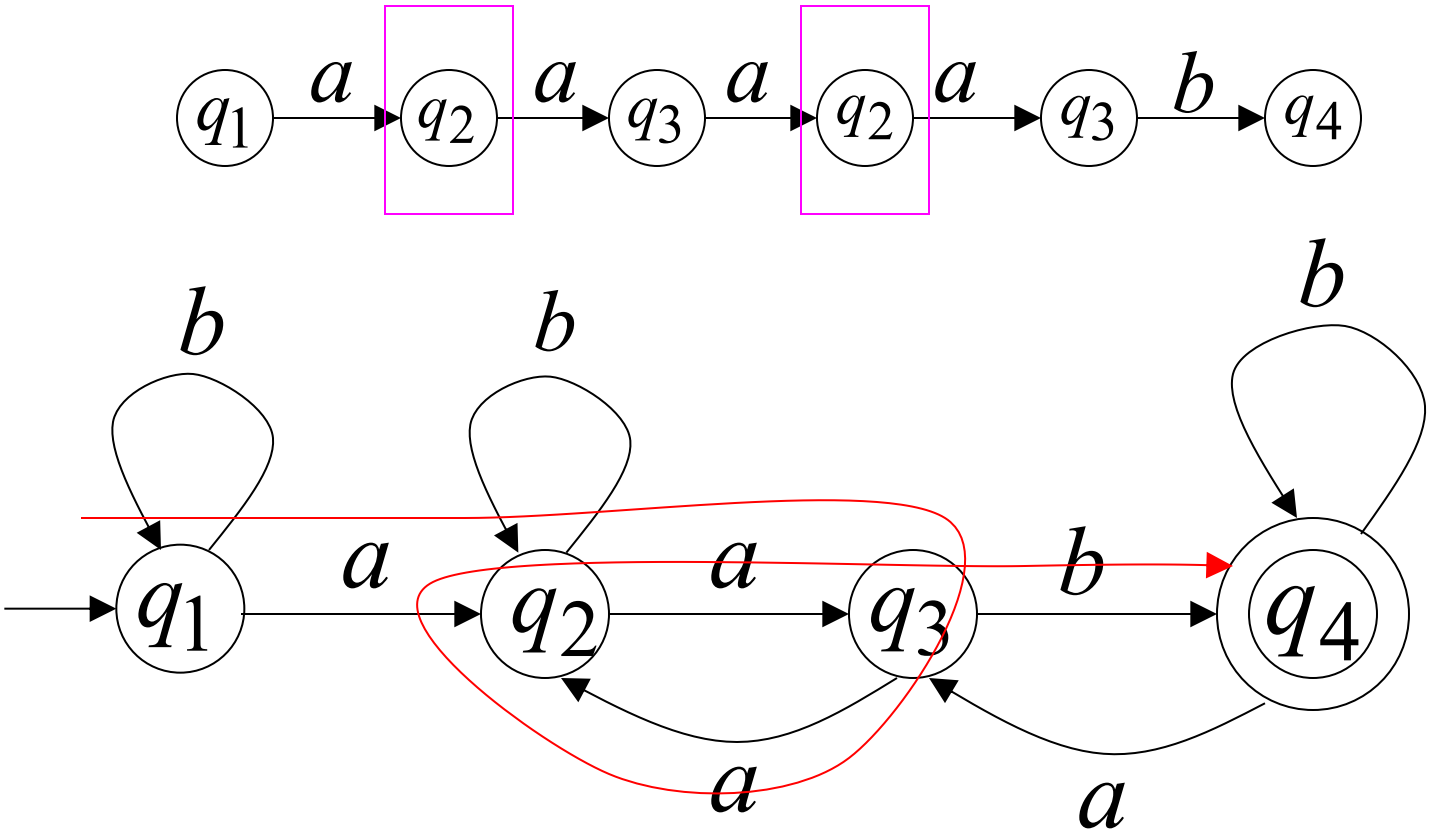
DFAs

Consider a DFA with 4 states



Consider the walk of a "long" string: $aaaab$ (length at least 4)

A state is repeated in the walk of $aaaab$

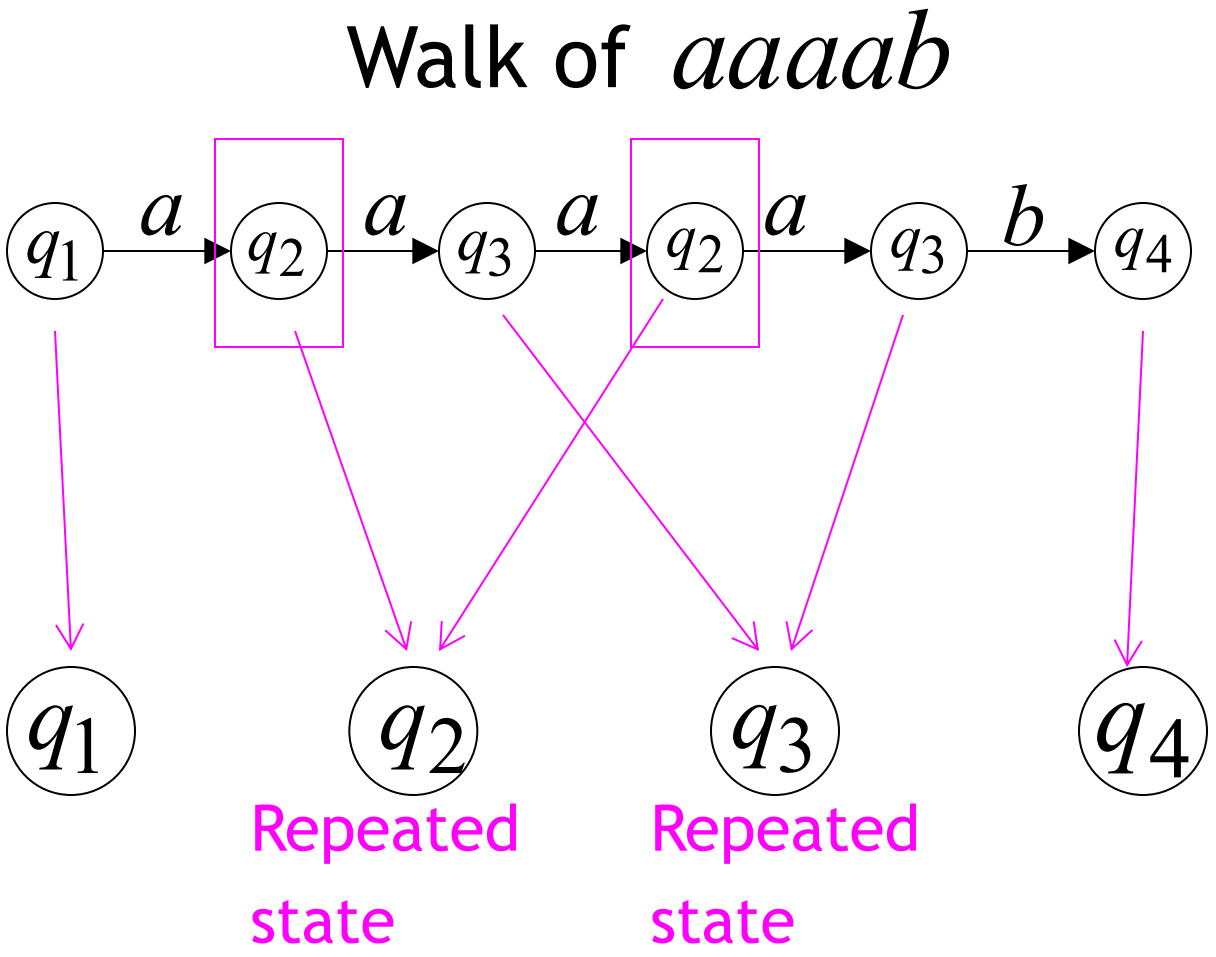


The state is repeated as a result of the pigeonhole principle.

Pigeons
(Walk states)

are more than

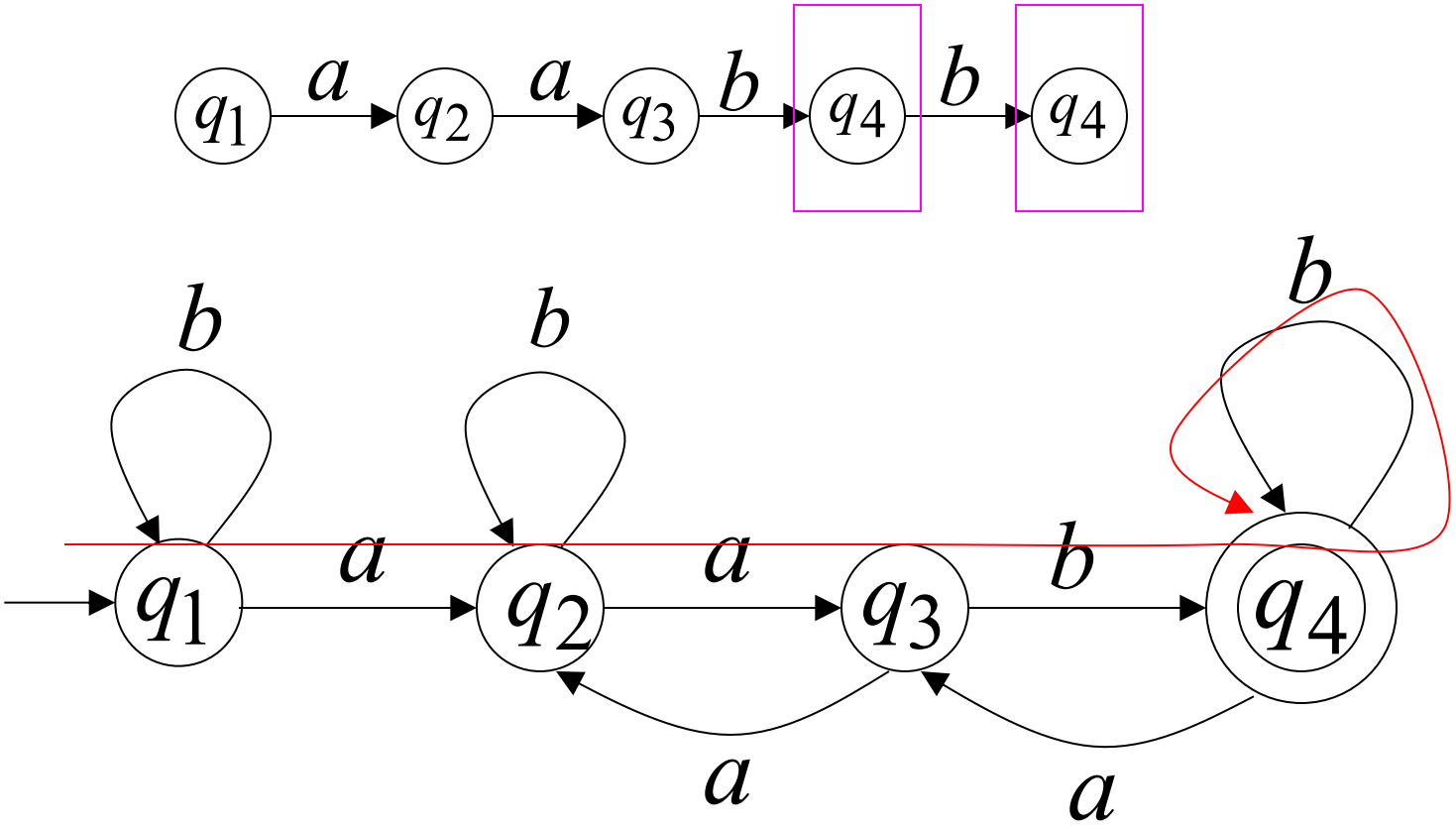
Nests
(Automaton states)



Consider the walk of a "long" string: $aabb$ (length at least 4)

Due to the pigeonhole principle:

A state is repeated in the walk of $aabb$

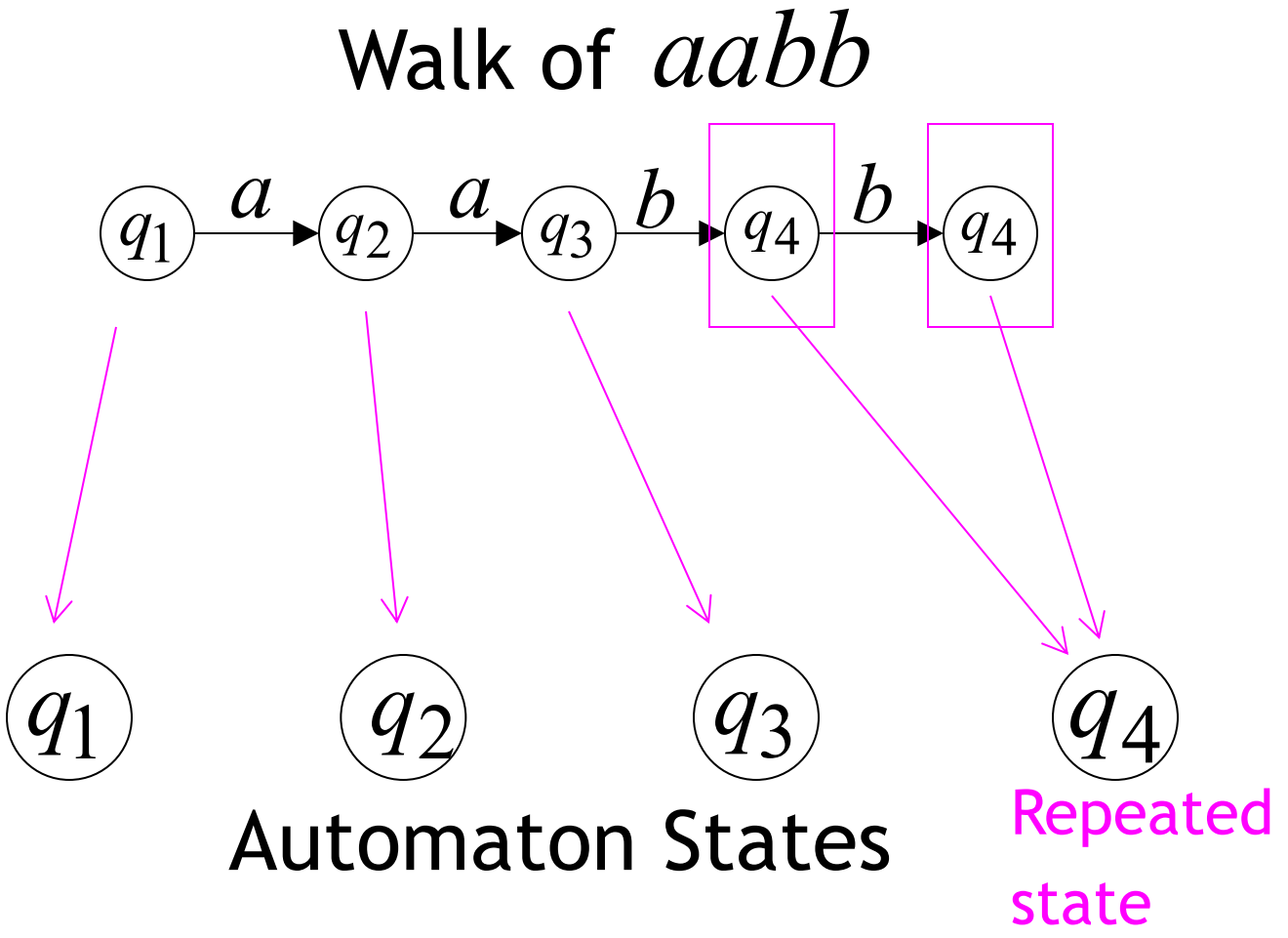


The state is repeated as a result of the pigeonhole principle:

Pigeons
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are more than

Nests
(Automaton states)

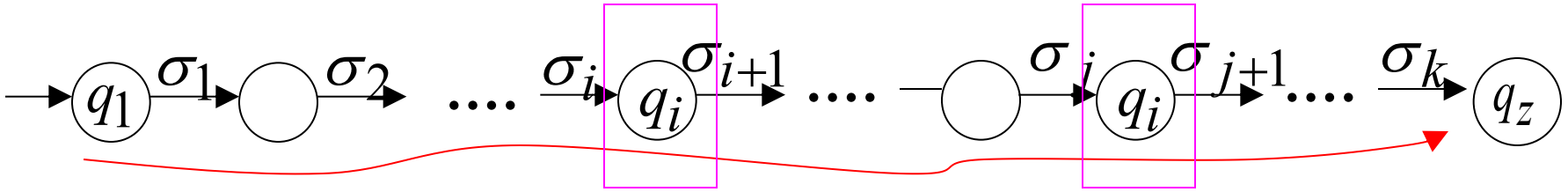


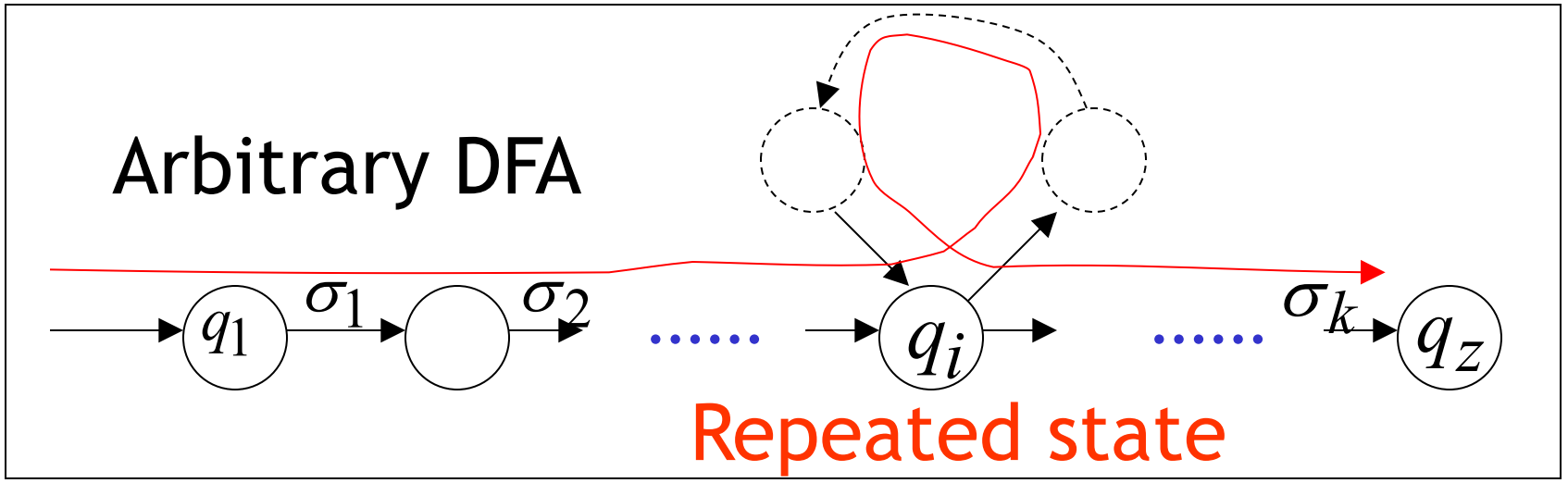
In General: If $|w| \geq \#\text{states of DFA}$,

by the pigeonhole principle,

a state is repeated in the walk w

Walk of $w = \sigma_1 \sigma_2 \cdots \sigma_k$



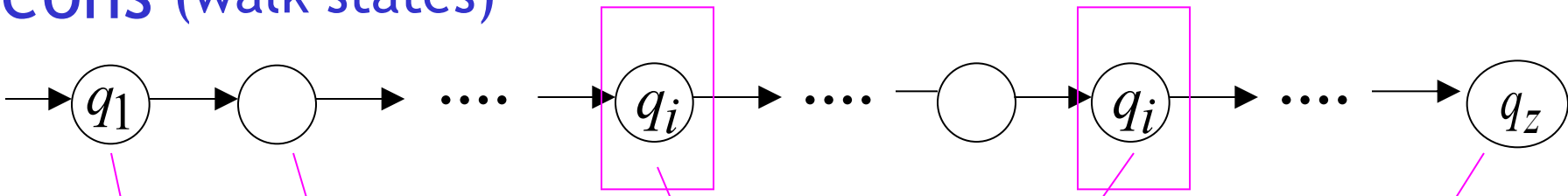


$$|w| \geq \#\text{states of DFA} = p$$

Number of states in walk is at least $p+1$

Walk of w

Pigeons (walk states)



are more than

Nests



(Automaton states)

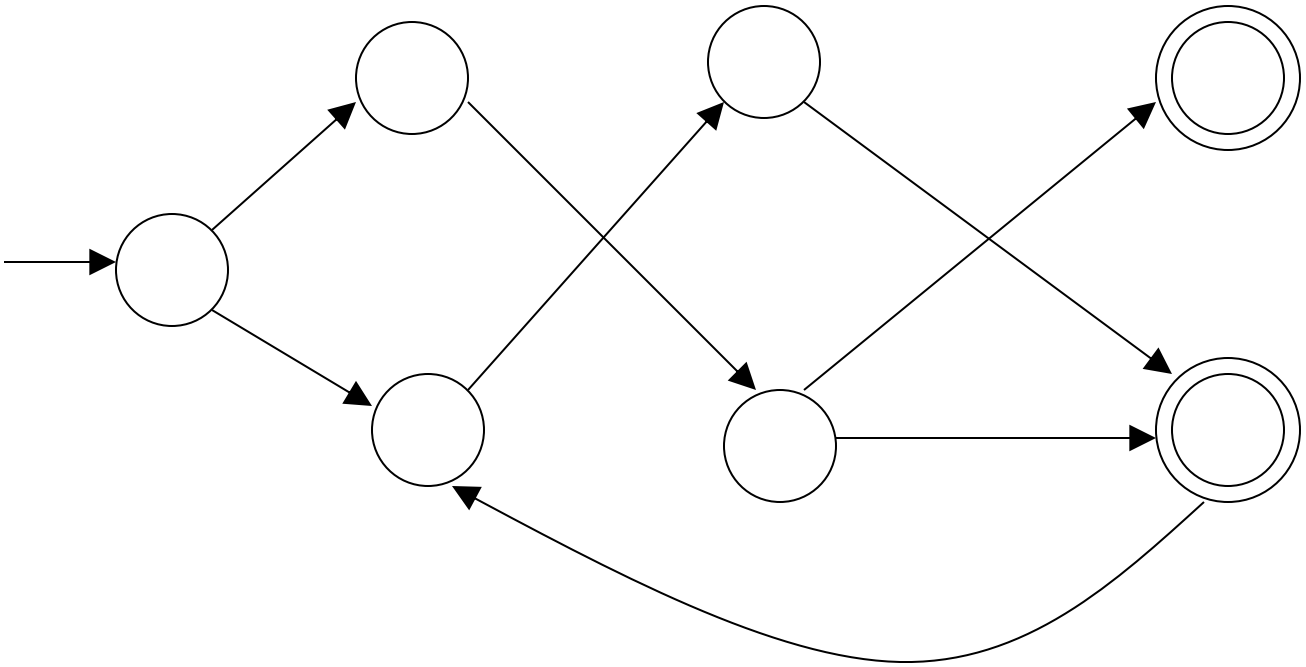
A state is repeated

The Pumping Lemma

Take an infinite regular language L

(contains an infinite number of strings)

There exists a DFA that accepts L

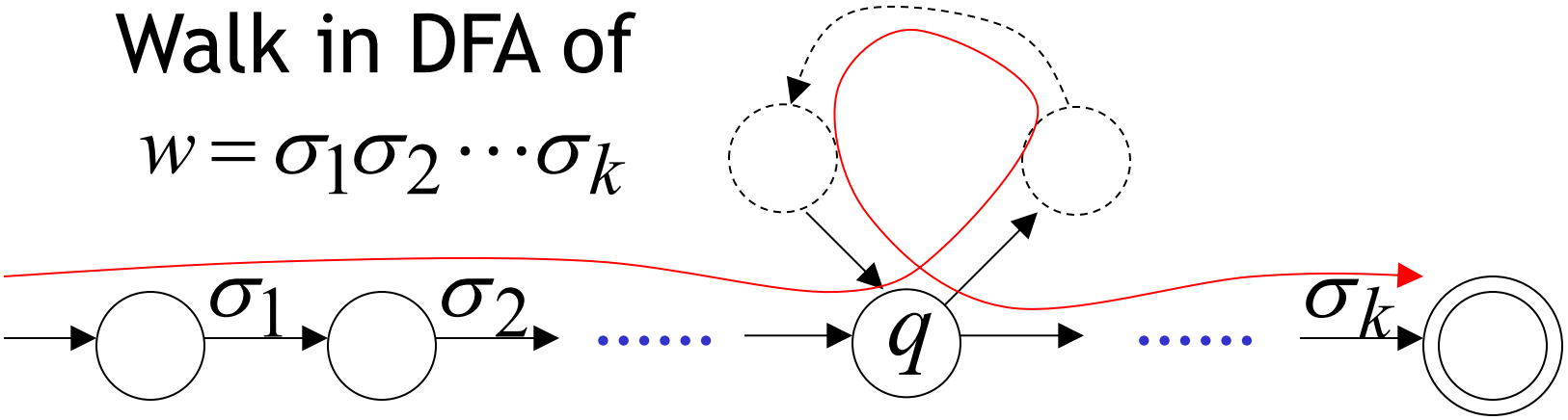


p states

Take string $w \in L$ with $|w| \geq p$

(number of states of DFA)

then, at least one state is repeated in the walk of w

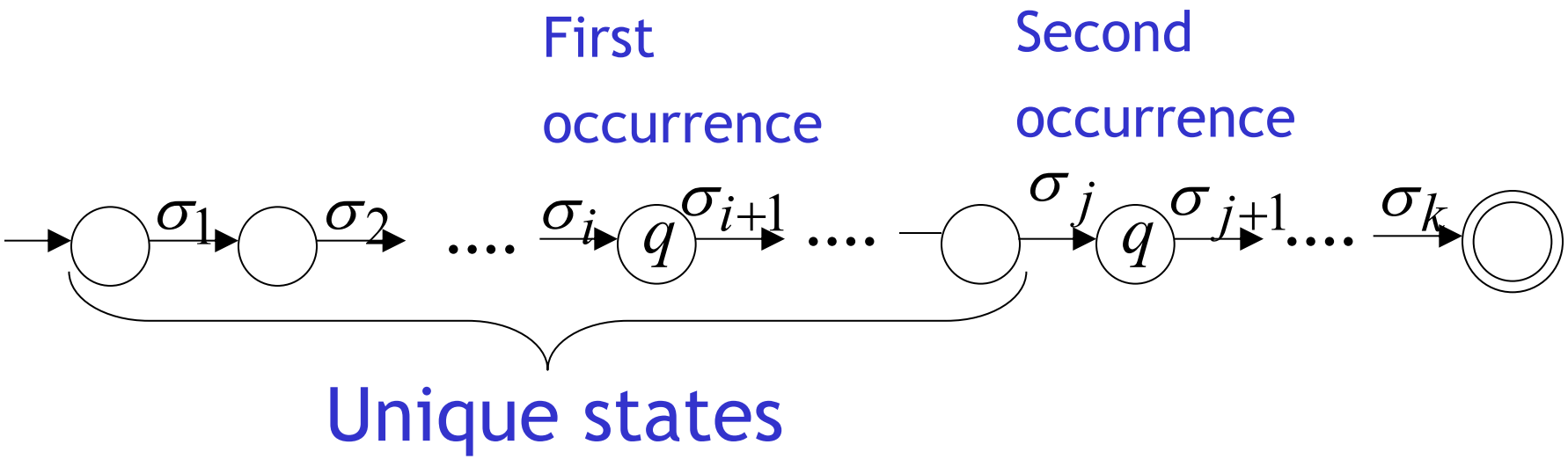


Repeated state in DFA

There could be many states repeated

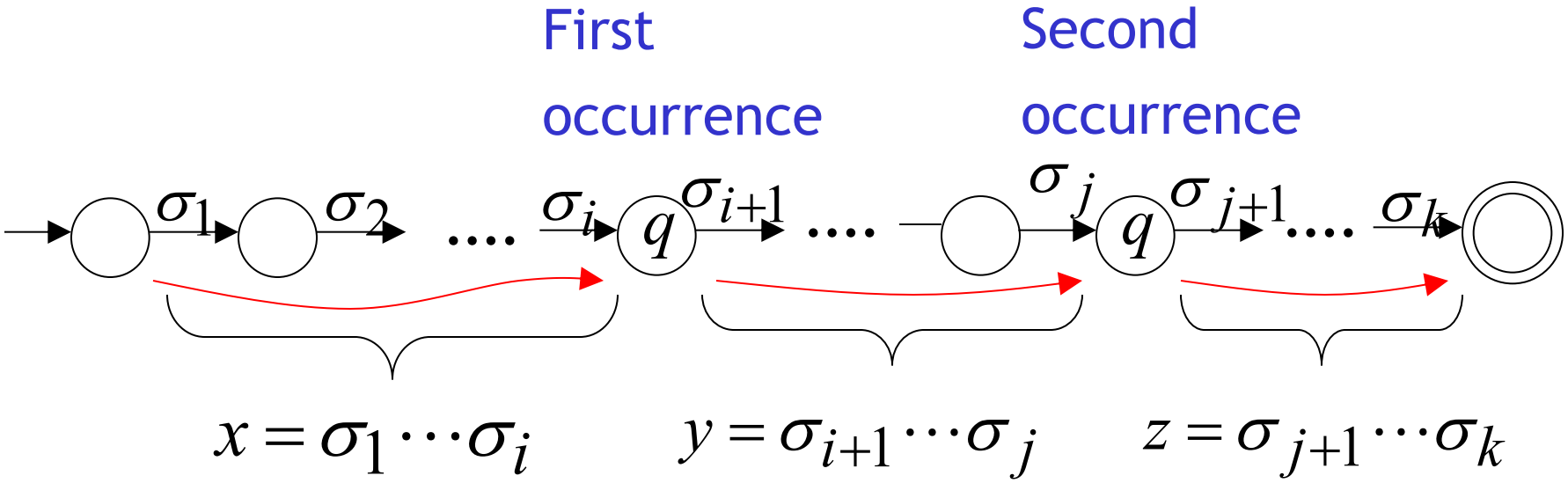
Take q to be the first state repeated

One dimensional projection of walk w :



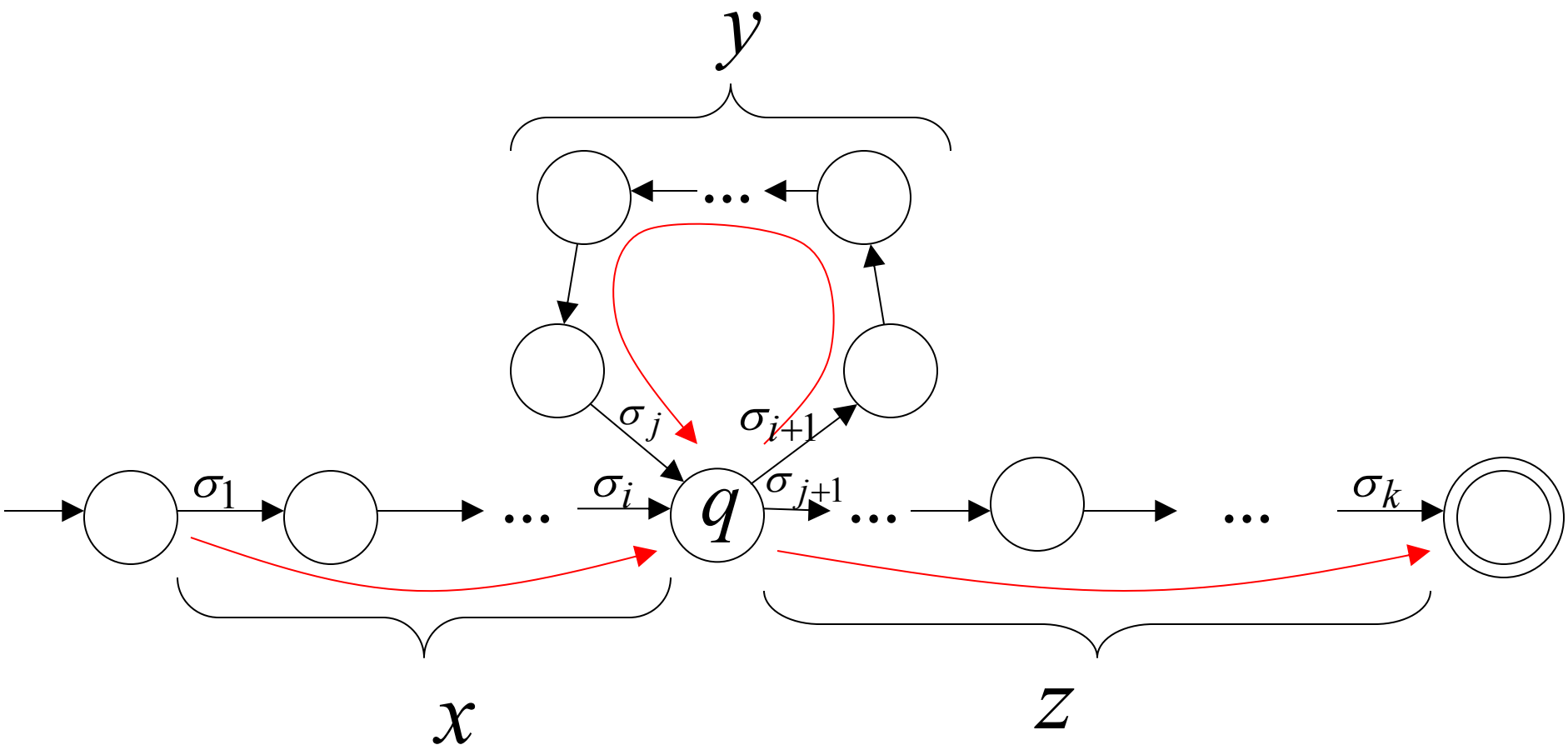
We can write $w = xyz$

One dimensional projection of walk w:

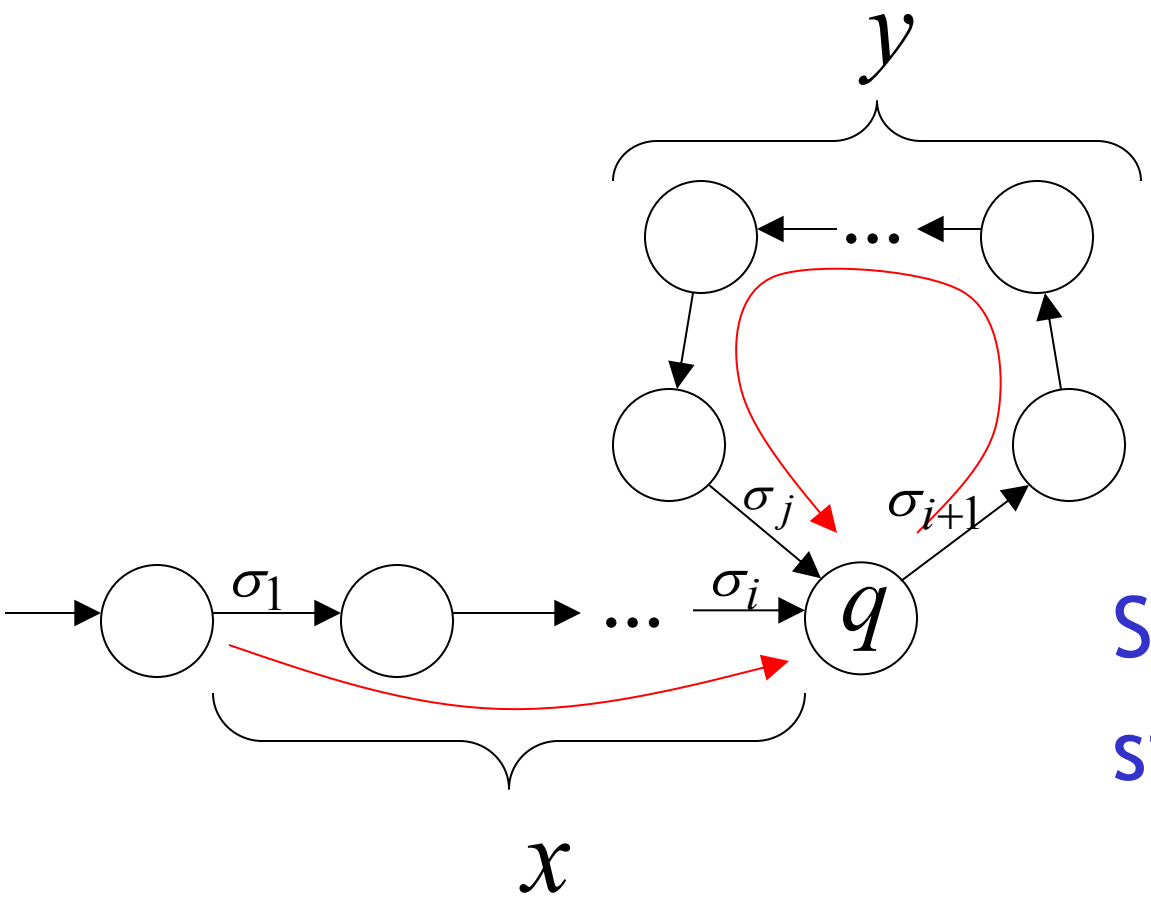


In DFA: $w = x\ y\ z$

Where y corresponds to substring between first and second occurrence of q



Observation: length $|xy| \leq p$ number of states of DFA



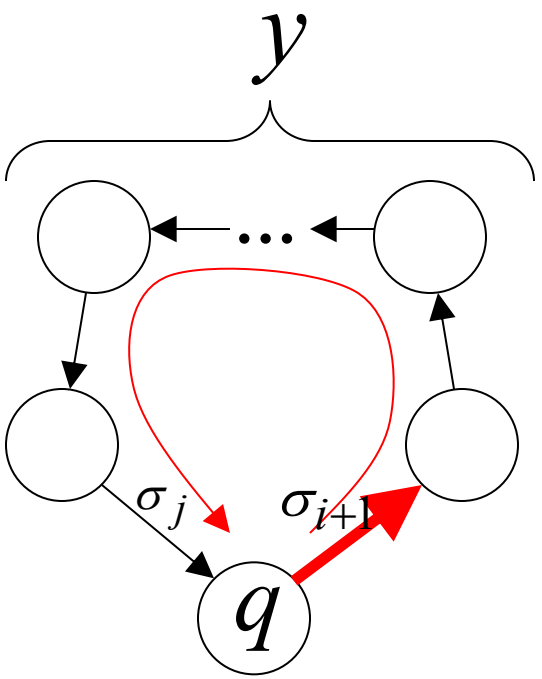
Because of unique states in xy

Since, in xy no state is repeated

(except q)

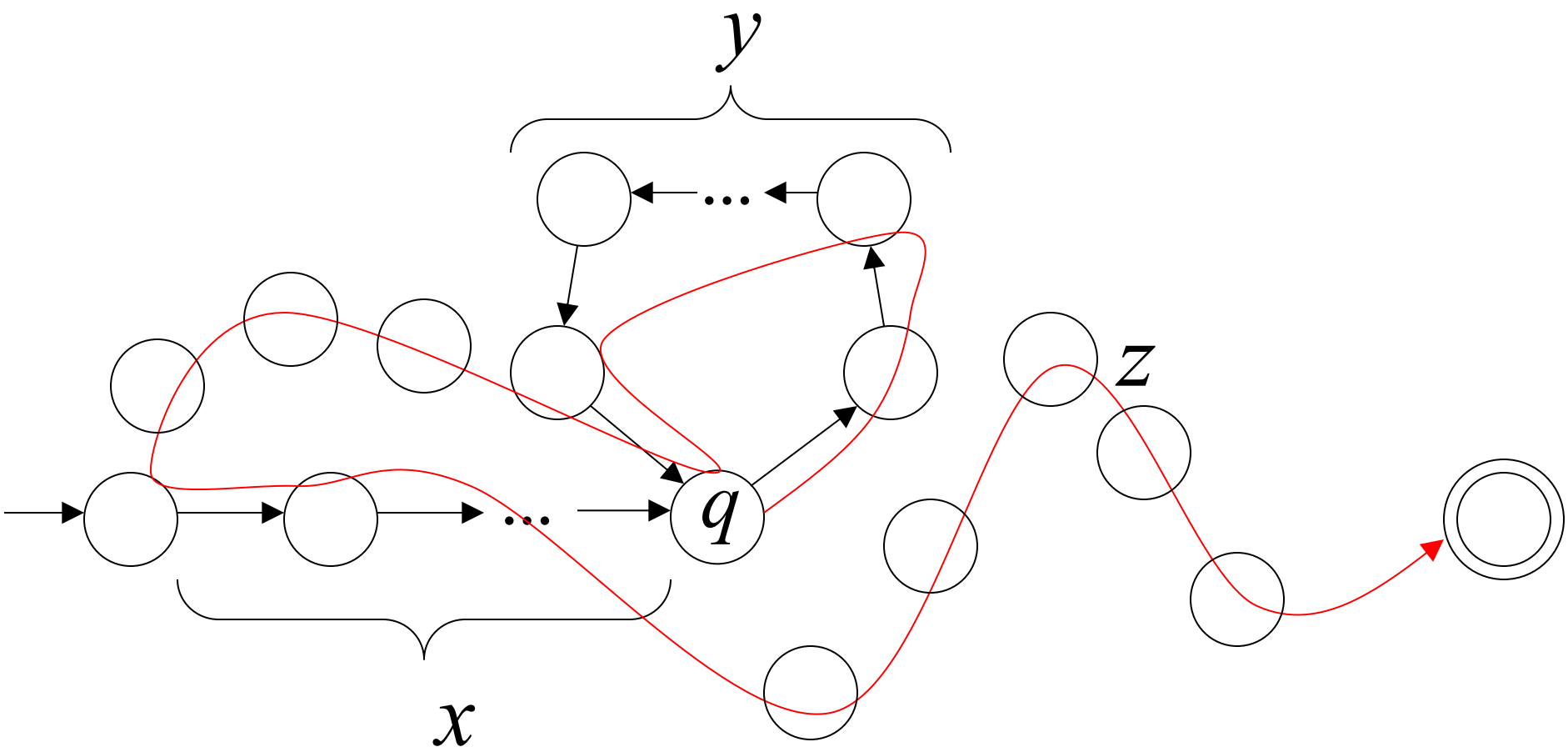
Observation: length $|y| \geq 1$

Since there is at least one transition in loop



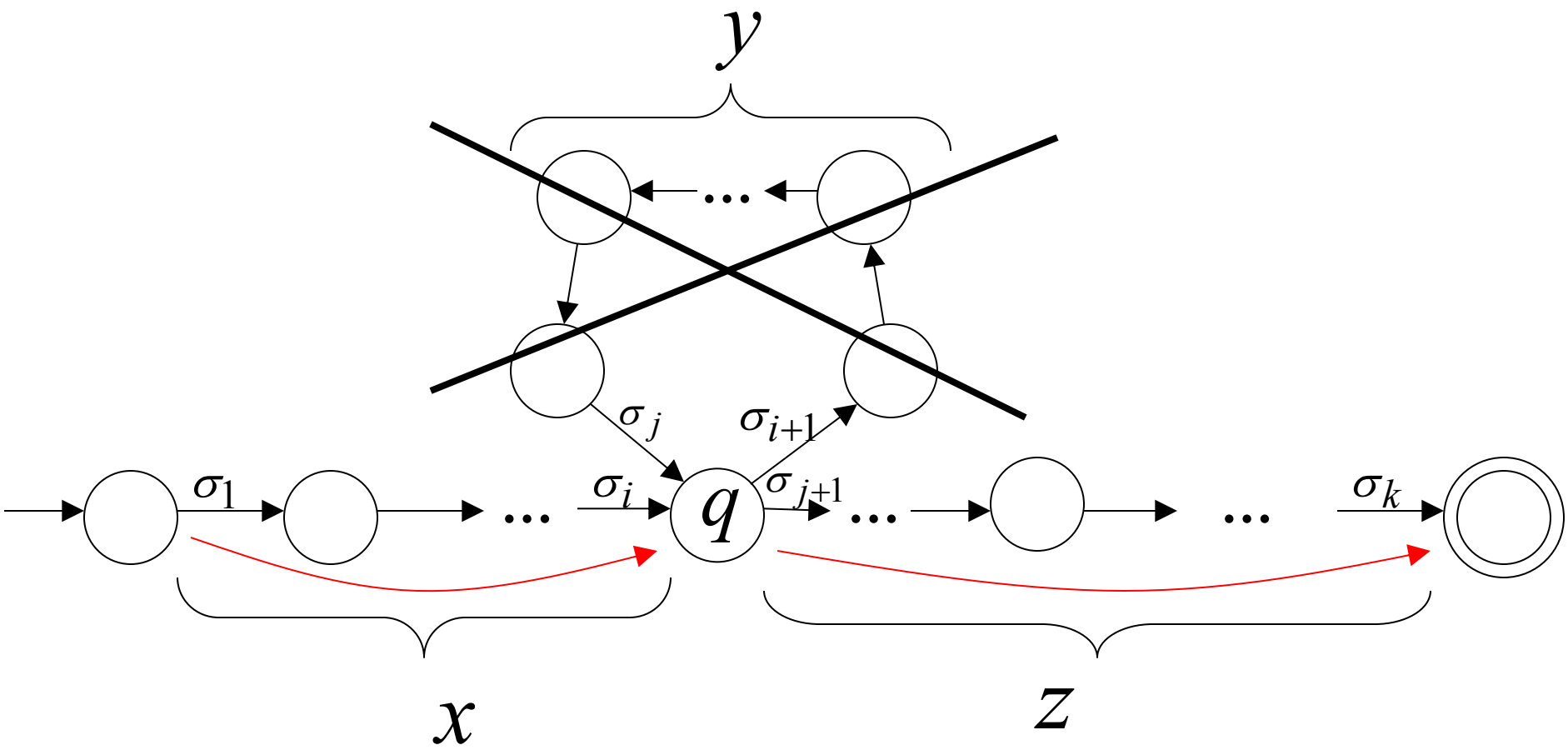
We do not care about the form of string z

z may actually overlap with the paths of x and y



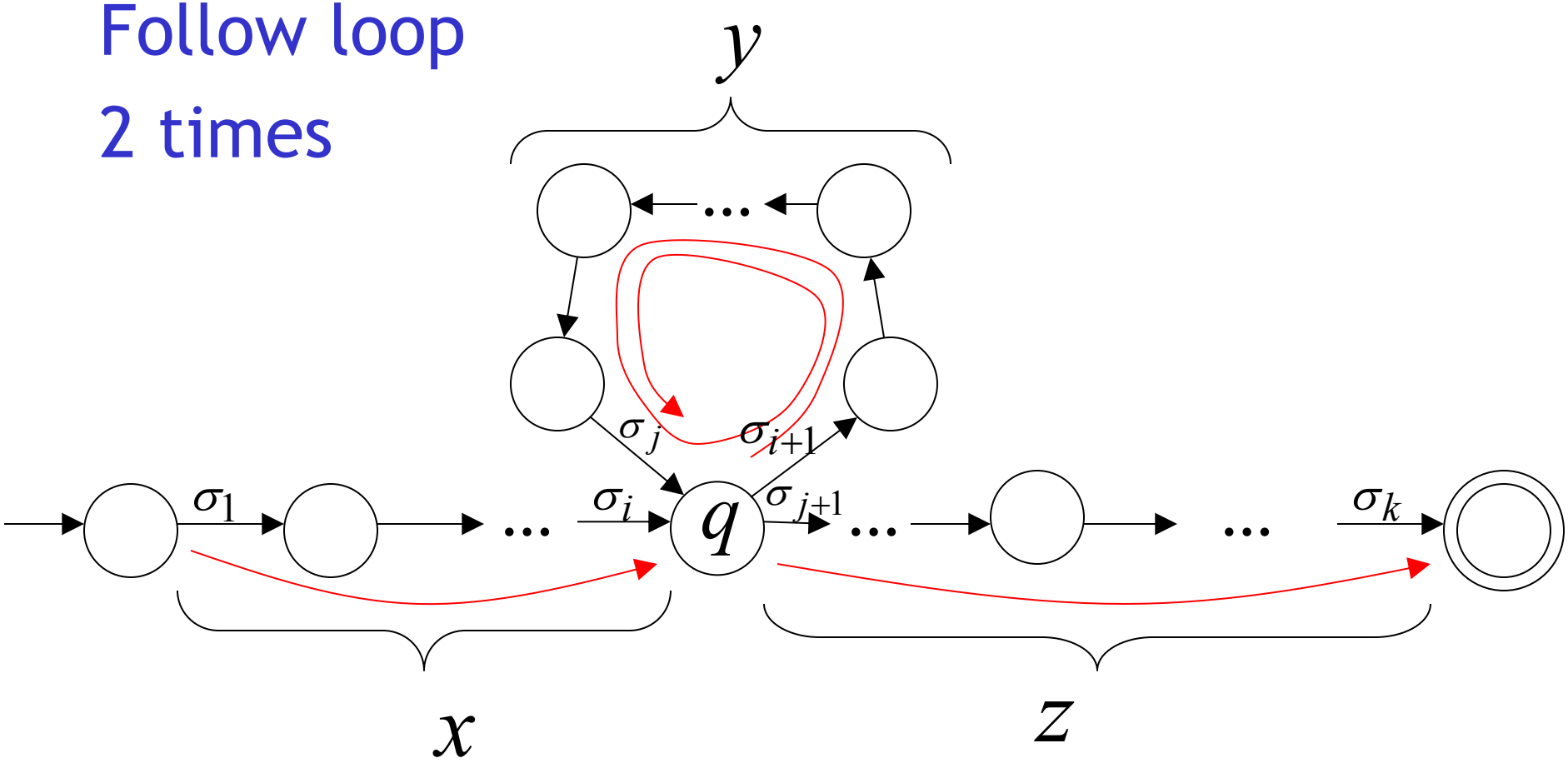
Additional string: The string $x\,z$ is accepted

Do not follow loop



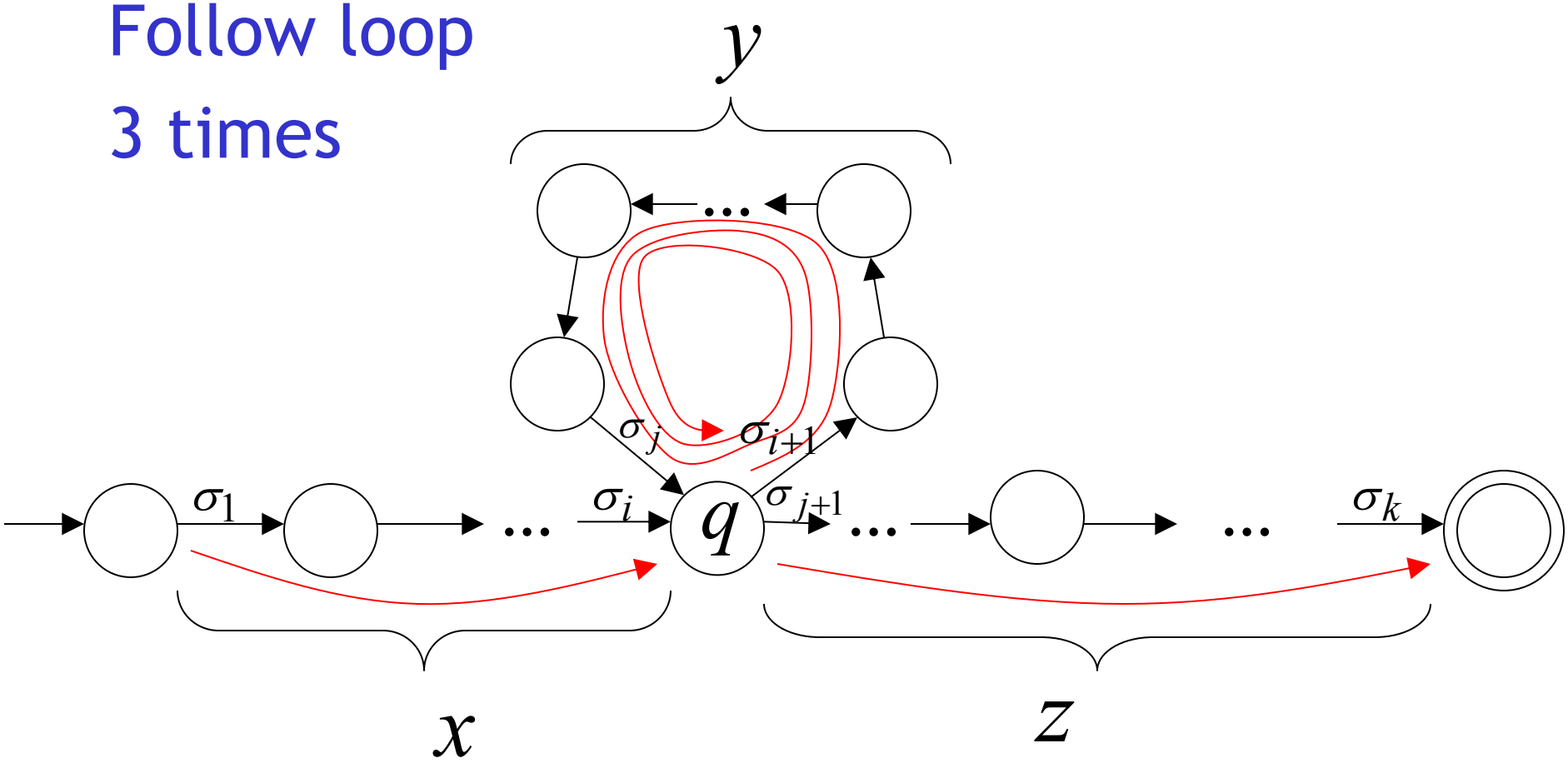
Additional string: The string $x\ y\ y\ z$ is accepted

Follow loop 2 times



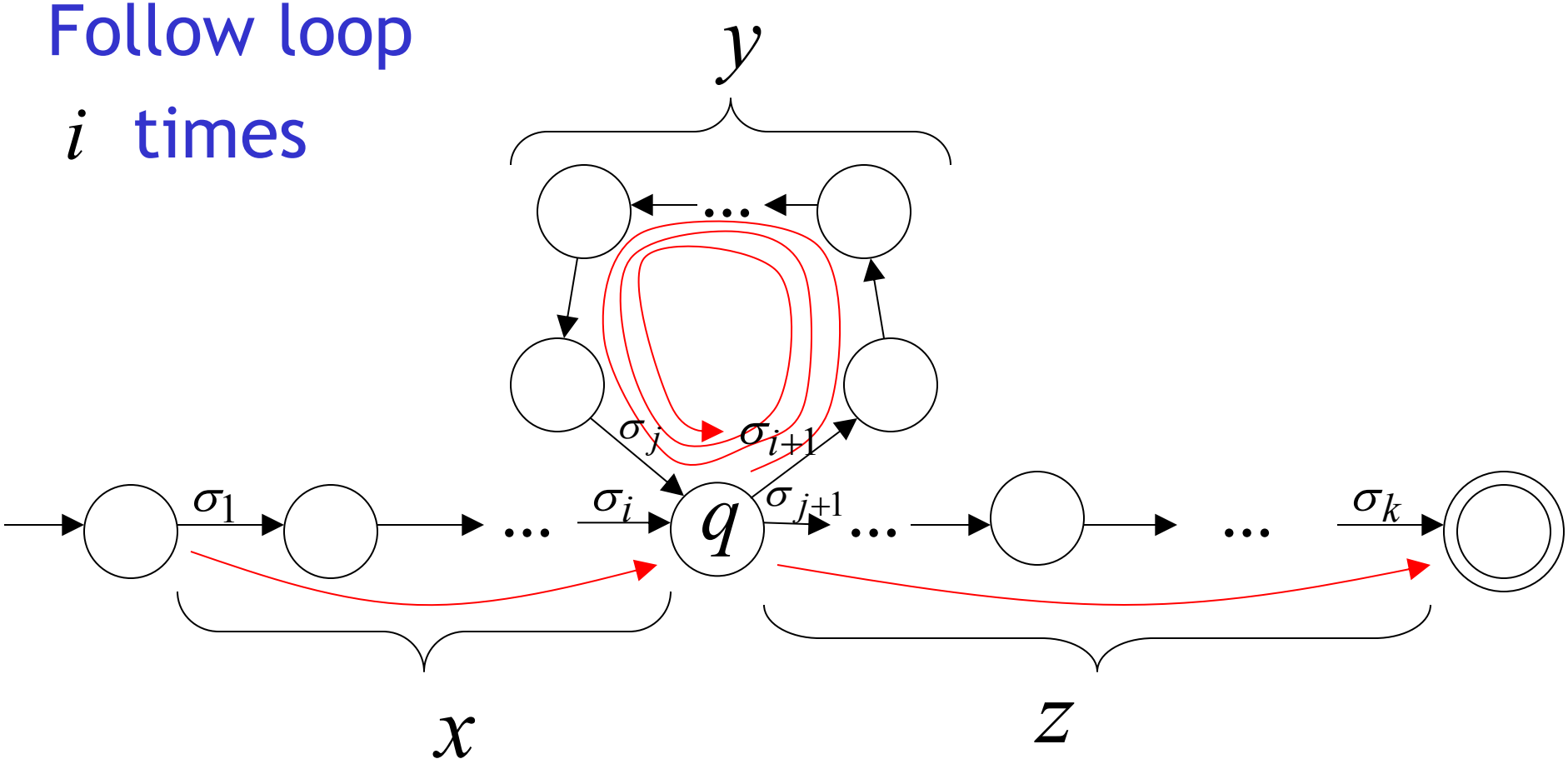
Additional string: The string $x\ y\ y\ y\ z$ is accepted

Follow loop 3 times



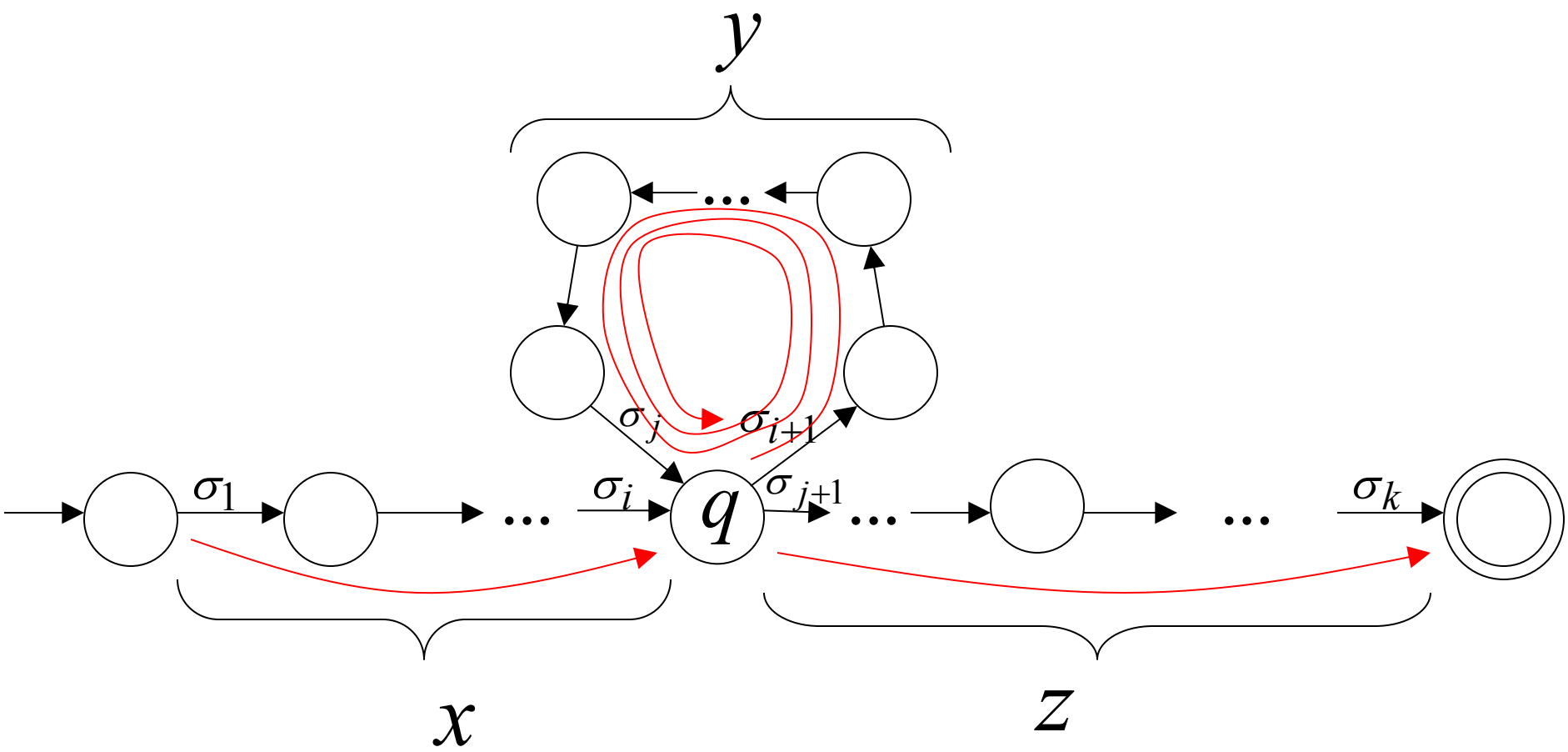
In General: The string $x\ y^i\ z \quad i = 0, 1, 2, \ldots$ is accepted

Follow loop i times



Therefore: $x\, y^i\, z \in L \qquad i = 0, 1, 2, \ldots$

Language accepted by the DFA



In other words, we described:

The Pumping Lemma!

The Pumping Lemma:

- Given a infinite regular language L
- there exists an integer p (critical length)
- for any string $w \in L$ with length $|w| \geq p$
- we can write $w = x\ y\ z$
- with $|x\ y| \leq p$ and $|y| \geq 1$
- such that: $x\ y^i\ z \in L \quad i = 0, 1, 2, \ldots$

Applications

of

the Pumping Lemma

Observation:

Every language of finite size has to be regular.

(We can easily construct an NFA
that accepts every string in the language.)

Therefore, every non-regular language
has to be of infinite size.

(contains an infinite number of strings)

Suppose you want to prove that
an infinite language L is not regular

1. Assume the opposite: L is regular.

2. The pumping lemma should hold for L.

3. Use the pumping lemma to obtain a contradiction.

4. Therefore, L is not regular.

Explanation of Step 3: How to get a contradiction

1. Let p be the critical length for L

2. Choose a particular string $w \in L$ which satisfies the length condition $|w| \geq p$

3. Write $w = xyz$

4. Show that $w' = xy^i z \notin L$ for some $i \neq 1$

5. This gives a contradiction, since from pumping lemma $w' = xy^i z \in L$

Note: It is sufficient to show that only one string $w \in L$ gives a contradiction.

You don't need to obtain contradiction for every $w \in L$.

Example Pumping Lemma Application

Theorem: The language $L = \{a^n b^n : n \geq 0\}$ is not regular

Proof: Use the Pumping Lemma

$$L = \{a^n b^n : n \geq 0\}$$

Assume for contradiction

that L is a regular language

Since L is infinite

we can apply the Pumping Lemma

$$L = \{a^n b^n : n \geq 0\}$$

Let p be the critical length for L

Pick a string w such that: $w \in L$

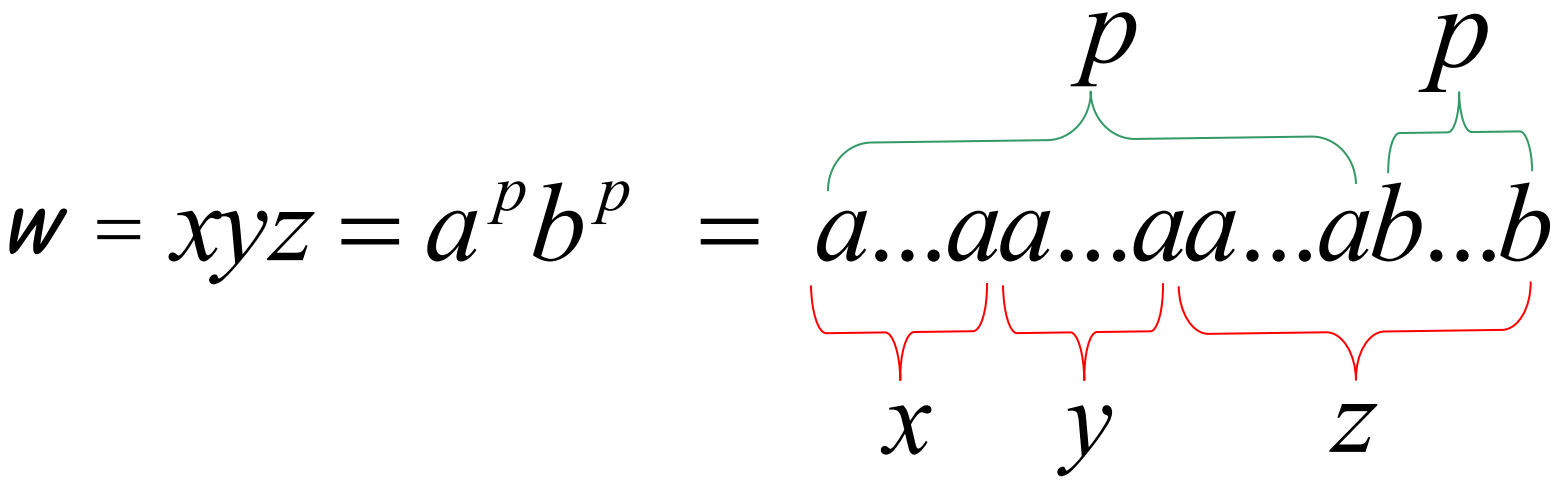
and length $|w| \geq p$

We pick $w = a^p b^p$

From the Pumping Lemma:

We can write $w = a^p b^p = x\ y\ z$

with lengths $|x\ y| \leq p, \quad |y| \geq 1$



Thus: $y = a^k, \quad 1 \leq k \leq p$

$$x\ y\ z = a^p b^p \qquad y = a^k,\ \ 1 \le k \le p$$

From the Pumping Lemma: $x\ y^i\ z \ \in\ L$

$$i = 0, 1, 2, \ldots$$

Thus: $x\ y^2\ z \ \in\ L$

$$x\ y\ z = a^p b^p \qquad y = a^k,\ \ 1 \le k \le p$$

From the Pumping Lemma: $x\ y^2\ z \ \in\ L$

$$xy^2z = \overbrace{\underbrace{a\ldots a}_{x}\underbrace{a\ldots a}_{y}\underbrace{a\ldots a}_{y}\underbrace{a\ldots a}_{z}}^{p+k}\overbrace{b\ldots b}^{p} \in L$$

Thus: $a^{p+k}b^p \in L$

$$a^{p+k}b^p \in L \qquad k \geq 1$$

BUT: $L = \{a^n b^n : n \geq 0\}$



$$a^{p+k}b^p \notin L$$

CONTRADICTION!

Therefore: Our assumption that L is a regular language is not true

Conclusion: L is not a regular language

END OF PROOF

Non-regular language $\{a^n b^n : \ n \geq 0\}$

